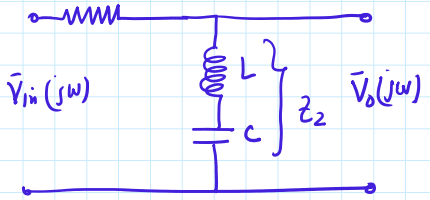
Band Rejection Filter:



$$\frac{\bar{V}_o}{\bar{V}_{in}}(j\omega) = T(j\omega) = \frac{\left(j\omega L - \frac{j}{\omega C}\right)}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

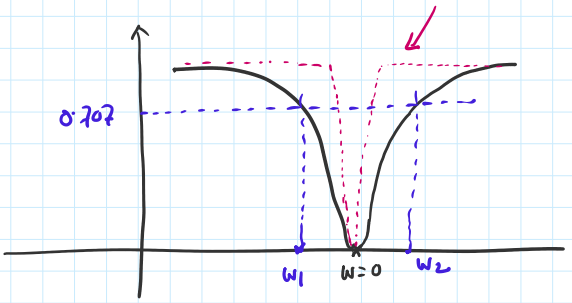
$$= \frac{j(\omega^2 LC - 1)/\omega C}{\omega RC + j(\omega^2 LC - 1)/\omega C}$$

$$= \frac{j(\omega^2 LC - 1)}{\omega RC + j(\omega^2 LC - 1)} = \frac{1}{1 - j\frac{\omega RC}{(\omega^2 LC - 1)}}$$

$$\bar{V}_o/\bar{V}_{in} = \frac{Z_2}{Z_1 + Z_2}$$

$$T(j\omega) = \frac{1}{\left[1 + \frac{(\omega RC)^2}{(\omega^2 LC - 1)^2}\right]^{1/2}} \Big/ \tan^{-1}\frac{\omega RC}{(\omega^2 LC - 1)}$$

$$= |T(j\omega)| \Big/ T(j\omega)$$



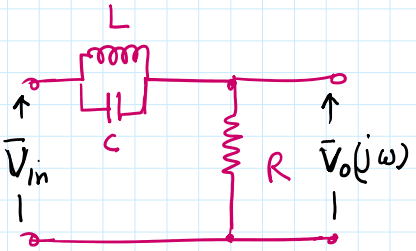
$$|T(j\omega)|_{\omega = \omega_0} = 0$$

$\omega < \omega_1$, $\omega > \omega_2$ } Signal is passed

$\omega_1 < \omega < \omega_2$ } signal is rejected } Band Rejection Filter

---

Another Circuit for Band



$$Z_L = j\omega L; \quad Z_C = \frac{1}{j\omega C}$$

$$Z_{LC} = Z_L \| Z_C = \frac{j\omega L \times \frac{1}{j\omega C}}{j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$= \frac{-j\,L/C}{(\omega^2 LC - 1)/\omega C} = \frac{j\omega L}{1 - \omega^2 LC}$$

Thus $$\frac{\bar{V}_o(j\omega)}{\bar{V}_{in}(j\omega)} = H(j\omega) = \frac{R}{R + Z_{LC}} = \frac{R}{R + \frac{j\omega L}{1 - \omega^2 LC}}$$

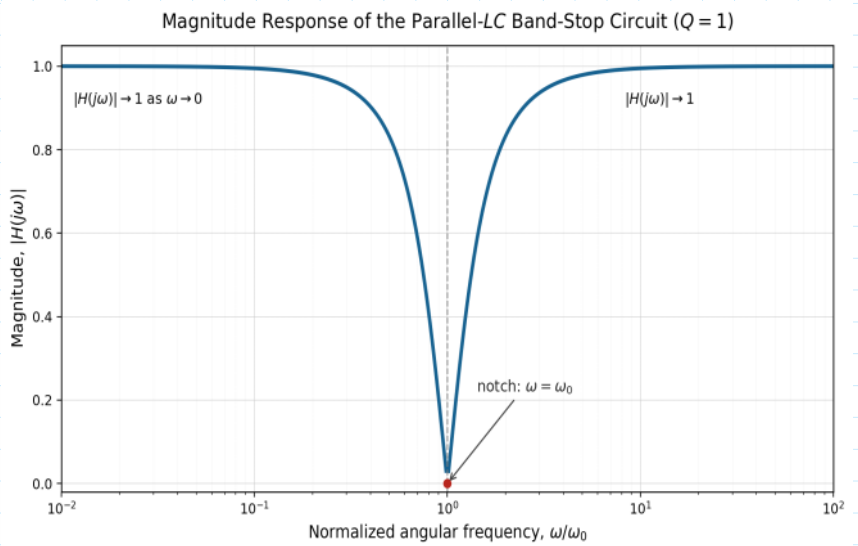
$$= \frac{R(1 - \omega^2 LC)}{\sqrt{R^2(1 - \omega^2 LC)^2 + (\omega L)^2}} = \frac{R\left\{1 - \left(\frac{\omega}{\omega_0}\right)^2\right\}}{\left[\quad\right]^{1/2}}$$

$$= \frac{R(1-\omega^2 LC)}{\sqrt{R^2(1-\omega^2 LC)^2 + (\omega L)^2}} = \frac{R\left\{1-\left(\frac{\omega}{\omega_0}\right)^2\right\}}{\left[R^2\left\{1-\left(\frac{\omega}{\omega_0}\right)^2\right\}^2 + (\omega L)^2\right]^{1/2}}$$

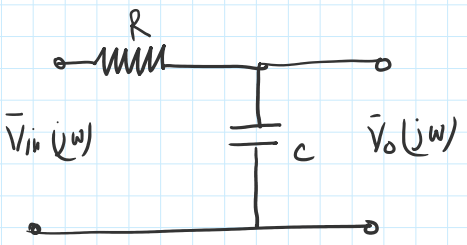
It can be seen that: when $\omega = \omega_0$, $|H(j\omega)| = 0$

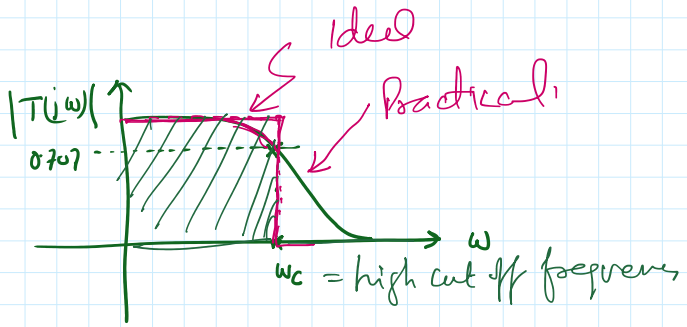
$\omega > \omega_0$ or $\omega < \omega_0$

$|H(j\omega)| \to 1$



Low Pass Filter.





$$\frac{\bar{V}_o(j\omega)}{\bar{V}_{in}(j\omega)} = \frac{(-j/\omega C)}{R + (-j/\omega C)} = \frac{1}{1 + j\omega RC} = \frac{1}{\sqrt{1+(\omega RC)^2}} \angle -\tan^{-1}(\omega RC)$$

$$= M \angle \phi$$

$$|T(j\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

$$|T(j\omega)|_{\omega=0} = 1$$

$$|T(j\omega)| = 0.707 = \frac{1}{\sqrt{1+(\omega RC)^2}} = \frac{1}{\sqrt{1+1}}$$

$$|T(j\omega)| = 0.707 = \frac{1}{\sqrt{1+(\omega RC)^2}} = \frac{1}{\sqrt{1+1}}$$

$$\Rightarrow \quad \omega_c RC = 1$$

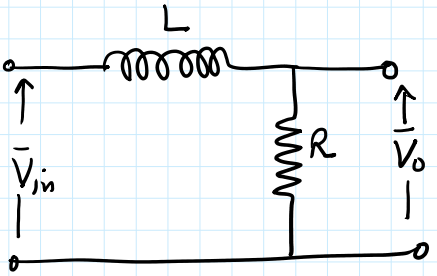
$$\boxed{\omega_c = \frac{1}{RC}} \quad \Rightarrow RC = \frac{1}{\omega_c}$$

Thus for given ω_c = higher cut-off frequency for LPF, we can find out R and C. This results in design of filter.

Another Circuit for Low Pass Filter:

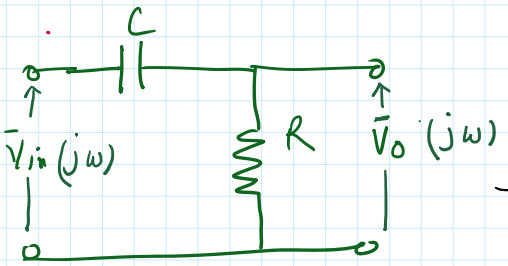
$$\left|\frac{\bar{V}_o}{\bar{V}_{in}}\right| = \left|\frac{R}{R+j\omega L}\right| = \frac{1}{\sqrt{1+(\omega\tau)^2}}$$

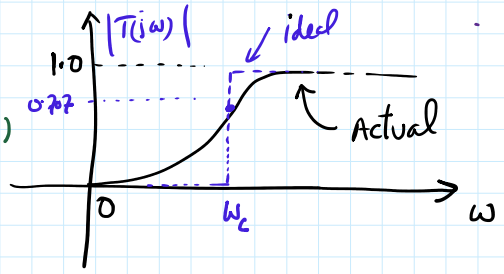
for $\tau = L/R$ = time constant of RL circuit.



The gain resembles Low pass filter characteristics.

High Pass Filter:





$$\frac{V_o(j\omega)}{V_{in}(j\omega)} = T(j\omega) = \frac{R}{R - j/\omega C}$$

$$= \frac{j\omega RC}{1+j\omega RC}$$

$$T(j\omega) = \left\{\frac{\omega RC}{\sqrt{1+(\omega RC)^2}}\right\} \angle 90^\circ - \tan^{-1}(\omega RC)$$

$$|T(j\omega)| = \frac{\omega RC}{\sqrt{1+(\omega RC)^2}}, \quad RC = \tau \text{ of RC circuit}$$

$$|T(j\omega)| = \frac{\omega RC}{\sqrt{1+(\omega RC)^2}} \quad ; RC = \tau \text{ of RC circuit}$$

$$= \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}}$$

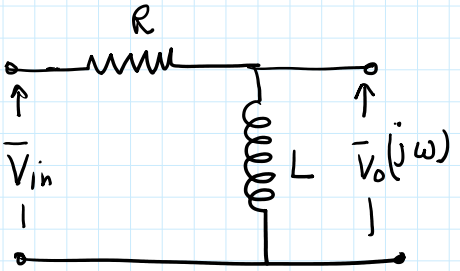
$$0.707 = \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}} = \frac{1}{\sqrt{2}}$$

$$\omega_c \tau = 1$$

$$\boxed{\omega_c = \frac{1}{RC}}$$

Another circuit for High Pass Filter:

$$\left| \frac{\bar{V}_o(j\omega)}{\bar{V}_{in}(j\omega)} \right| = \frac{\omega L}{\sqrt{R^2+(\omega L)^2}}$$



$$= \frac{\omega\tau}{\sqrt{1+(\omega\tau)^2}}$$

which has high pass filter characteristics.

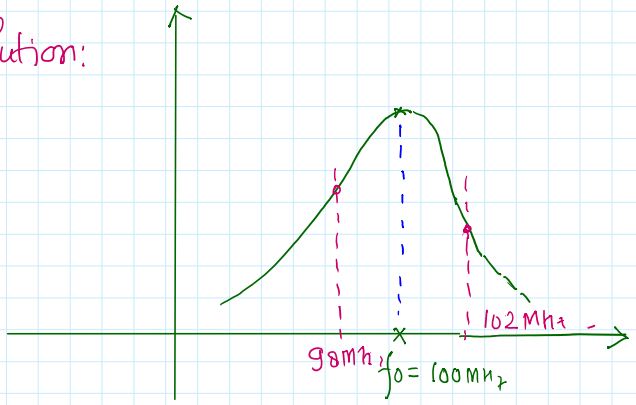
Question:

A university student is designing a simple FM receiver to tune to the campus radio station at 100 MHz. Two nearby stations operate at 98 MHz and 102 MHz. The receiver front-end uses a series RLC band-pass filter with the output taken across a 50 Ω resistor. The filter should provide maximum response at 100 MHz and attenuate each neighbouring station by at least 20 dB relative to the desired station.

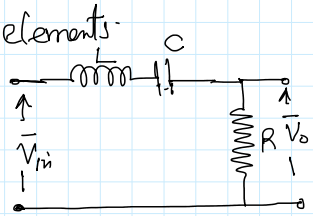
Using only a 50 Ω resistor, a variable inductor, and a variable capacitor, design the filter and determine:

(a) the minimum required quality factor Q,
(b) the corresponding values of L and C, and
(c) comment briefly on whether the obtained component values are practically realizable for an FM receiver.

Solution:



Band Pass Filter: with R, L and C elements.



$$\left|\frac{\bar{V}_o}{\bar{V}_{in}}\right| = \left|\frac{1}{1+j\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)}\right|$$

$$= \left|\frac{1}{1+jQ_0\left(\frac{2\pi f}{2\pi f_0}-\frac{2\pi f_0}{2\pi f}\right)}\right|$$

$$= \frac{1}{\sqrt{1+Q_0^2\left(\frac{f}{f_0}-\frac{f_0}{f}\right)^2}}$$

At resonance $f = f_0$

Gain in decibels $= 20\log\left|\frac{\bar{V}_o}{\bar{V}_{in}}\right| = -20$

$$\Rightarrow \left|\frac{V_o}{V_{in}}\right| = 10^{-1} = 0.1$$

Thus $$\frac{1}{\sqrt{1+Q_0^2\,(f/f_0 - f_0/f)^2}} = 0.1$$

$$\Rightarrow 1+Q_0^2(f/f_0 - f_0/f)^2 = \left(\frac{1}{0.1}\right)^2 = 100$$

$$\Rightarrow Q_0^2(f/f_0 - f_0/f)^2 = 99$$

$$Q_0 = \frac{\sqrt{99}}{|(f/f_0 - f_0/f)|}$$

Now lower adjacent station: 98 MHz

$$\Rightarrow Q_{0L} = \frac{\sqrt{99}}{\left|\frac{98}{100}-\frac{100}{99}\right|} = 246.2$$

$Q_{0L} \geq 246.2$

Upper Adjacent station: 98 MHz

$$Q_{0H} = \frac{\sqrt{99}}{\left|\frac{102}{100}-\frac{100}{102}\right|} = 251.2$$

Thus $\boxed{Q_0 = 251.2}$ as it provides better attenuation at lower freq. station

Now $\frac{\omega_0 L}{R} = 251.2$; $\omega_0 = 2\pi f_0 = 2\pi \times 100\times10^6$

$R = 50\,\Omega$, $L = \frac{251.2\times 50}{2\pi\times100\times10^6} = 19.99\times10^{-6}$

$$R = 50\,\Omega, \quad L = \frac{251.2 \times 50}{2\pi \times 100 \times 10^6} = 19.99 \times 10^{-6}$$

$$\boxed{L \simeq 20\ \mu H}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = 2\pi \times 100 \times 10^6$$

$$\Rightarrow \sqrt{LC} = \frac{1}{2\pi \times 10^8}$$

$$= \frac{1}{(2\pi \times 10^8)^2 \times 19.99 \times 10^{-6}}$$

$$C = 126.7\,pF = 127\,pF$$

$$\boxed{C = 127\ pF}$$

Practical Observation

Although the calculated inductance and capacitance are within conventional component ranges, achieving a loaded quality factor of about 251 at 100 MHz using a simple lumped series RLC circuit is challenging. Real inductors exhibit winding resistance, parasitic capacitance, and a finite self-resonant frequency. In a practical FM receiver, a higher-order RF filter or an appropriately designed tuned/distributed network would normally be preferred when substantial adjacent-channel rejection is required.